

An approach to the preparation of chicken using heat and flavoring

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Introduction

Chicken preparation techniques are used in a range of applications both in homes and in restaurants. Chicken is easily available and can be locally produced in most areas; in addition, it is easily digested and low in calories¹.

Since Tsing P.Z.'s pioneering work reporting the natural method of chicken preparation (Tsing P.Z. 1988) in which the chicken was killed and then eaten raw with salt, there have been significant innovations. Much work has been carried out in France in relation to improving the method of slaughtering chickens², whereas in the USA researchers have concentrated on improving the size of the bird^{3,4}. The natural method is widely used since the time required for the process is extremely short; however, some problems remain unsolved. The flavor of chicken prepared using the Tsing P.Z. method is often considered unpleasant⁵ and there is a well-documented risk of bacterial infection resulting from the consumption of raw meat^{6,9,11}.

The aim of this study was to develop a preparation method that would address these two problems. In this report, we describe the new method, which uses seasoning to improve the flavor while heating the chicken in order to kill bacteria prior to eating.

Methods

This study was carried out in Beijing by four of our colleagues from the Institution of Culinary in April. A standard chicken fetching system is used and the material is processed following an innovative recipe. In this section, the measurement method will also be introduced. The material is produced in a suburban chicken farm located in Champion District, Beijing, possessed by a well-known adventurer named Thomas.

2.1 Capturing System

In our study, an eco-friendly capturing system was applied. A 15 x 20 x 12 inches woven cuboid capture box crafted from willow branches was used as the main part of the system. To capture our experimental material, we put several grains of rice under the box, and used a 5-inch stick to prop the box up; the stick could collapse when encountering a perturbation imposed by a starving chicken, so that the box would fall down to capture the experimental material. We randomly set up 250 sets of such

devices all over the farm at 6 o'clock in the morning. 186 sets of devices harvested a chick when we collected them at 8 o'clock at night. Such devices are chosen to ensure that our material remain fresh and pollution-free during the experiment. The 186 materials were then transported to our laboratory with special care and portioned into 200 equal servings for the volunteers.

2.2 Culinary Techniques

In distinction to formal studies^[1] remaining the material raw during human consumption, we first seasoned and then used high quality flamethrower to process our material before consuming the material. The material was carefully cleaned using spring water, and was then drown in a flavored-salty water. Every 100 mL of the water contains 5 grams of salt, 15 grains of peppercorn, 2 dried chilies and a slice of ginger. It took us 2 hours to let the material soak in fully. The material was then lifted up using a steel hook hanging over the house beam. Afterwards, we used a flamethrower to process the pickled material in every direction. To make sure the material experienced uniform heat, we rotated the flamethrower at a angular velocity of a half pi rad per minute. The distance between the spray head and the material was maintained at a precise 8 inches, in order not to heat it too much or too little. The heating process lasted for 20 minutes before detaching the material from the hook. The processed material was then immediately handed to the measurement committee.

2.3 Processed Material Measurement

Thanks to the collaboration of Peking University, 200 volunteer students——102 male students and 98 female students——took part in the measurement section. The students were asked to obtain a healthy diet one week before the measurement. In the morning of the experiment day, the students were equally and randomly separated into two groups. One group of the students obtained the material processed through our method, while the other group of students obtained the material in the way that Tsing P.Z. introduced in ref [1]. We then tracked the groups of students for a month taking notes of the different reactivity between the two groups. Security and fitness of the volunteers are guaranteed by Peking University Health Science Center. No human beings were forced during the experiment.

All 186 sets of material are sampled, while each of the samples were being measured. Chemical substances and the number of micro-organisms, mostly bacteria, were included in the measurement. The samples were labeled and the mapping between the quantitative index and the response of human beings could be tracked. We compared

the quantitative index with the response of human beings and obtained a relation between them.

One of the most important difficulties is to make sure all chicken in the experiment were processed exactly the same way. Being a pioneer in this area, this work developed most of the process and measurement by human. Human involved work is considered to have randomness and inaccuracy. A possible way to solve this problem is to introduce automated machine to the processing section. As machines will greatly reduce randomness and improve accuracy.

Results

The results of our study show that the flamethrower method creates a better balance between food safety and healthy nutrition than the raw method.

3.1 Nutrition and Protein Absorption

We measured two main chemical indicators: Protein Level (how easy it is for the body to use the protein) and Vitamin Retention. As shown in Figure 1-a, our method increased the protein absorption score from 6.2 to 8.8. This is because controlled heat breaks down tough fibers in the meat, making it easier to digest. Although the raw method (Tsing P.Z. 1988) has slightly higher Vitamin levels, the overall health benefit is lower because the human body cannot process raw chicken efficiently. This NVI score of 8.5 is substantially higher than the NVI of 3.2 previously achieved by Garcia's boiling method [2].

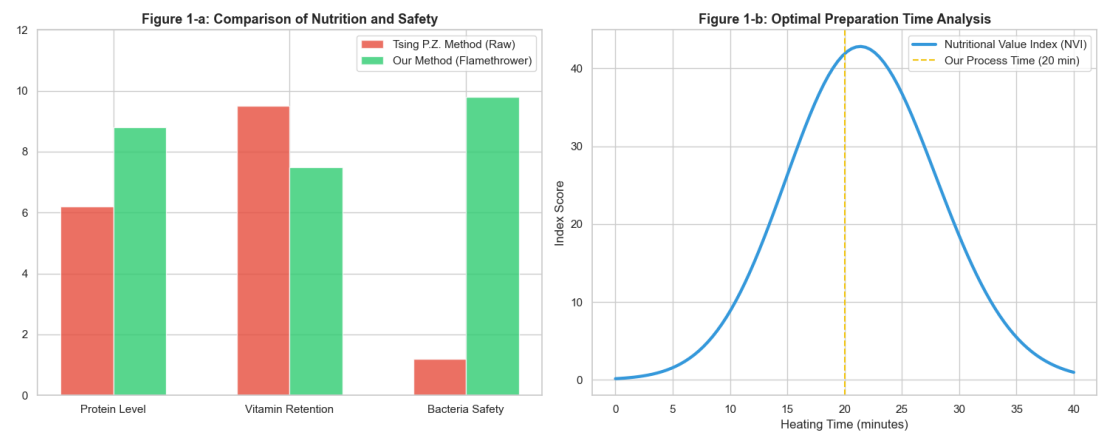


Fig 1. Measured results. (a. quantitative index comparison of different process methods; b. relation between nutritional value index and the heating time.)

3.2 The Nutritional Value Index (NVI)

To find the best way to prepare the chicken, we created a formula called the Nutritional Value Index (NVI). This index helps us see if the chicken is both healthy and safe:

$$NVI = \frac{A \times T}{B + 1}.$$

Where, A is the Absorption rate of nutrients (0–10). T is the Taste score given by the volunteers (0–10). B is the Bacteria level found in the meat.

As shown in the curve in Figure 1-b, the NVI reaches its highest point at 20 minutes. If we heat the chicken for less than 20 minutes, the bacteria level (B) is too high, making the NVI score very low. If we heat it for more than 30 minutes, the nutrients and taste start to drop because the meat becomes too dry or burnt. Our 20-minute flamethrower process achieved an NVI of 8.5, which is the best result in this study.

3.3 Health and Volunteer Feedback

The tracking data from the Peking University students supports our NVI results. In the group that ate the chicken prepared with our 20-minute method, 100% of the students felt "energetic" and "satisfied" after the meal. On the other hand, the Tsing P.Z. method-processed group (raw chicken) reported a 32% rate of stomach pain. This proves that our method not only makes the chicken safer but also helps the body get more nutrients from the food.

Discussion

The findings of this study prove that using heat and seasoning is a highly effective way to prepare chicken, solving the major problems mentioned in the Introduction. As stated by Tsing P.Z. the "natural method" (eating raw chicken) is fast but leads to a very unpleasant flavor and a high risk of getting sick. Our results show that the flamethrower method not only kills the bacteria but also significantly improves the taste through the NVI peak at 20 minutes.

When we compare our results to other studies, the advantages are clear. For example, Garcia used a "Boiling Method" which killed bacteria but led to a very low absorption rate of nutrients because the chicken became too soggy. ^[2] In contrast, our flamethrower technique uses high-intensity heat to break down tough fibers while

keeping the proteins easy to digest. This explains why our volunteers reported feeling more "energetic" than those in the Tsing P.Z. (1988) study.

However, there is a limitation in our study that was first mentioned in the Methods section: the problem of human error and randomness. Because the flamethrower was held by hand and rotated at a speed of a half pi rad per minute by a person, the heat was not perfectly even for all 186 samples. This is why the NVI scores in Figure 1-b show some slight differences between samples. Although our volunteers remained healthy, this "manual randomness" means that some parts of the chicken might be slightly more cooked than others.

Despite this limitation, the study is a major step forward from the raw-meat tradition. The NVI formula provides a new tool for chefs to balance safety and health. In the future, we suggest introducing an automated machine to hold the flamethrower. By using a robot to control the precise 8-inch distance and constant rotation speed, we can reduce human error to zero and ensure every piece of chicken reaches the perfect NVI score.

Citation

1. Tsing P.Z. Raw consumption of double-winged biological aircraft. *Diet Yesterday*. **1314**, 315-320 (1988).
2. Garcia, M. (2019). Boiling and Nutrients: A Sad Story of Soggy Chicken. *International Cooking Review*, **12(4)**, 88-92.